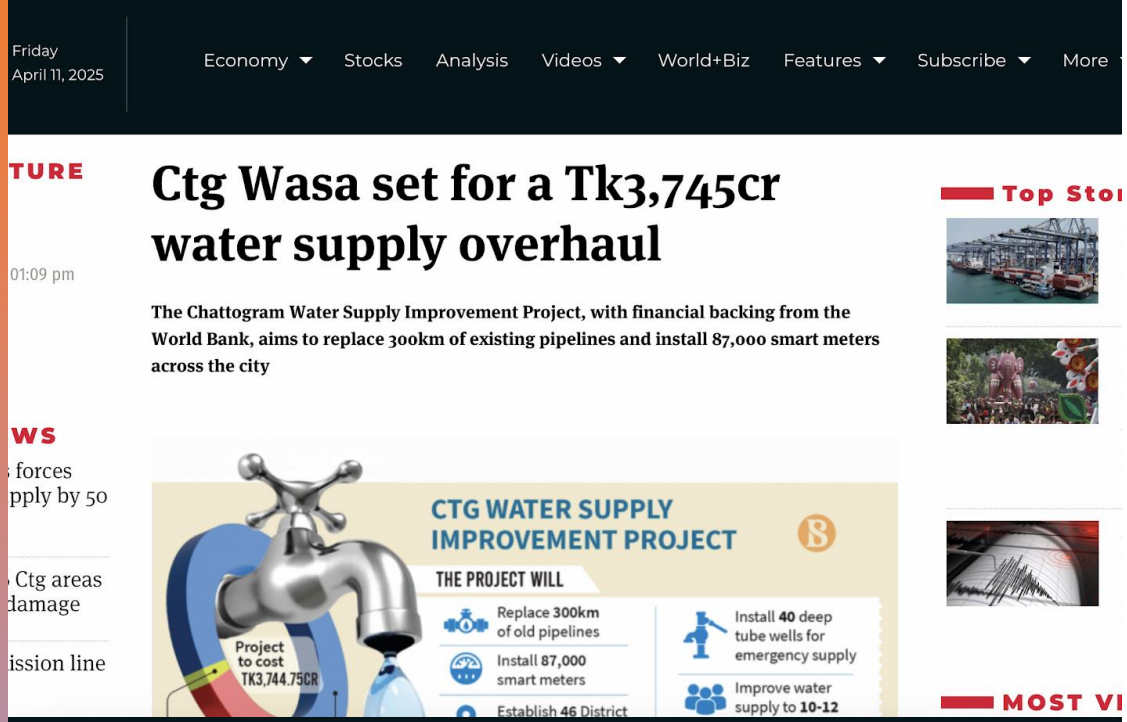


CLIX: AI-Driven Smart Utility Management System for Bangladesh

Presented by: CodeCrafters



THE PROBLEM - CRITICAL INFRASTRUCTURE LOSSES





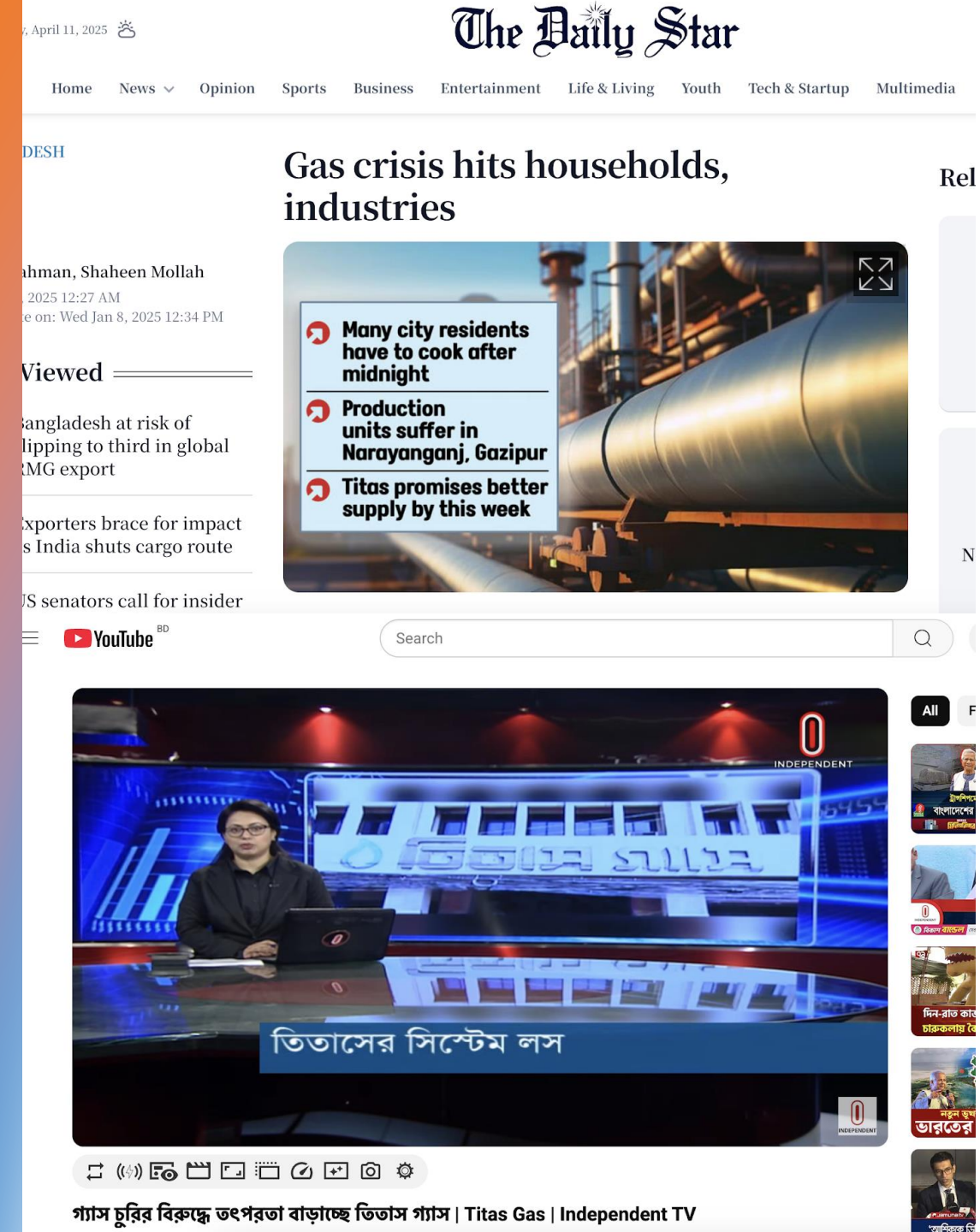
Water Supply Crisis

- "Chattogram WASA experiences 25-30% non-revenue water loss, much higher than the global standard of less than 5%" (The Business Standard, 2024)
- "A loss of Tk14-17 crore worth of water every month" (The Business Standard, 2024)
- "42 Meter Inspectors for 78,000 connections" leads to inaccurate readings (The Business Standard, 2022)



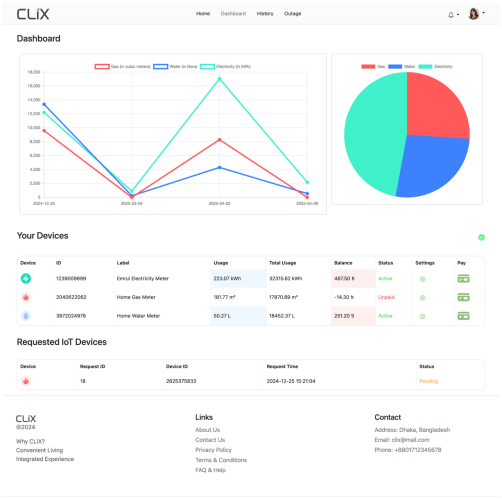
Electricity Distribution Failures

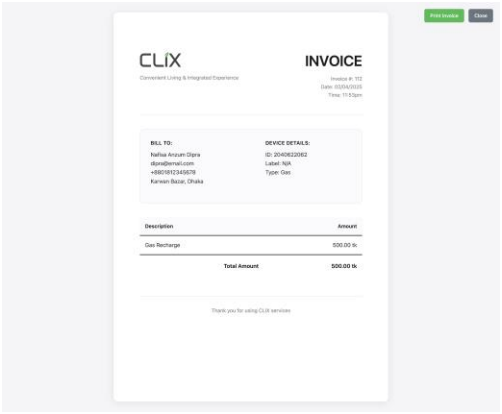
- "5 lakh defective meters out of 3.5 crore" nationwide (ATN News - Jul 16, 2024)
- Elderly users struggle with "token codes with over 100 digits" (CLIX exclusive interview)
- "Four people were electrocuted when the power cable fell into the waterlogged area in Mirpur" (DhakaTribune, 2023)

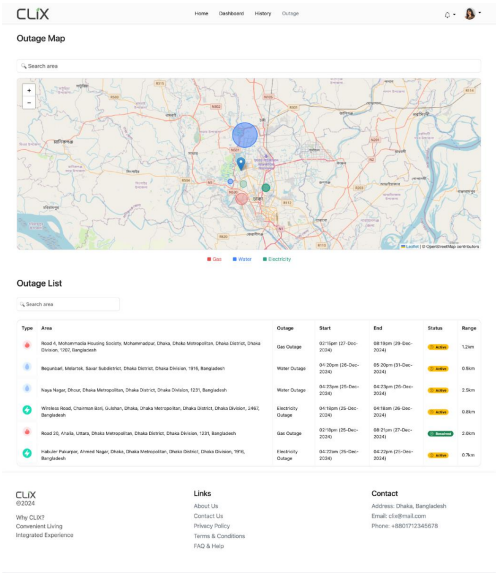


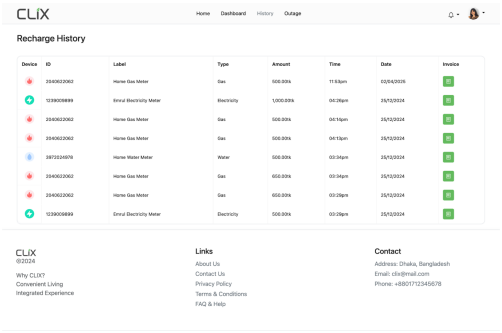
Gas System Inefficiencies

- Titas Gas reported system losses of "7.67%, where BRC standardized 2%" (Independent TV, 2025)
- Losses costing "3000 Cr BDT annually" (Independent TV, 2025)
- Factories "cutting production by almost a third" due to inadequate gas pressure (Daily Star, 2025)







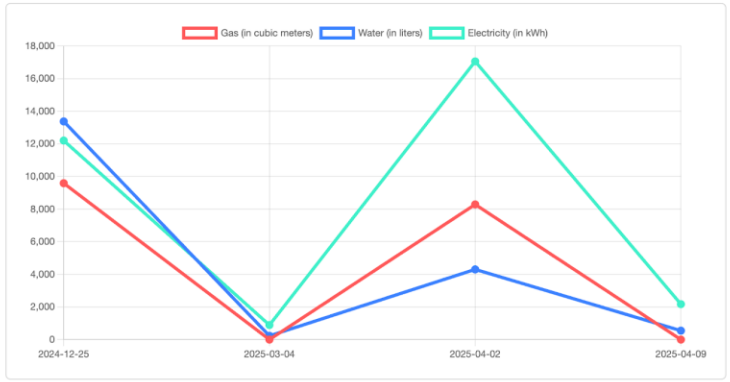


The Solution: CLIX (Convenient Living & Integrated Experience)

CLIX

Home Dashboard History Outage

Dashboard





Your Devices

Device	ID	Label	Usage	Total Usage	Balance	Status	Settings	Pay
	1239009899	Emrul Electricity Meter	223.07 kWh	32315.62 kWh	467.50 ৳	Active		
	2040622062	Home Gas Meter	181.77 m³	17870.89 m³	-14.30 ৳	Unpaid		
	3972024978	Home Water Meter	50.27 L	18452.37 L	251.20 ৳	Active		

Requested IoT Devices

Device	Request ID	Device ID	Request Time	Status
	18	2625375833	2024-12-25 15:21:04	Pending

CLIX
©2024

Why CLIX?
Convenient Living
Integrated Experience

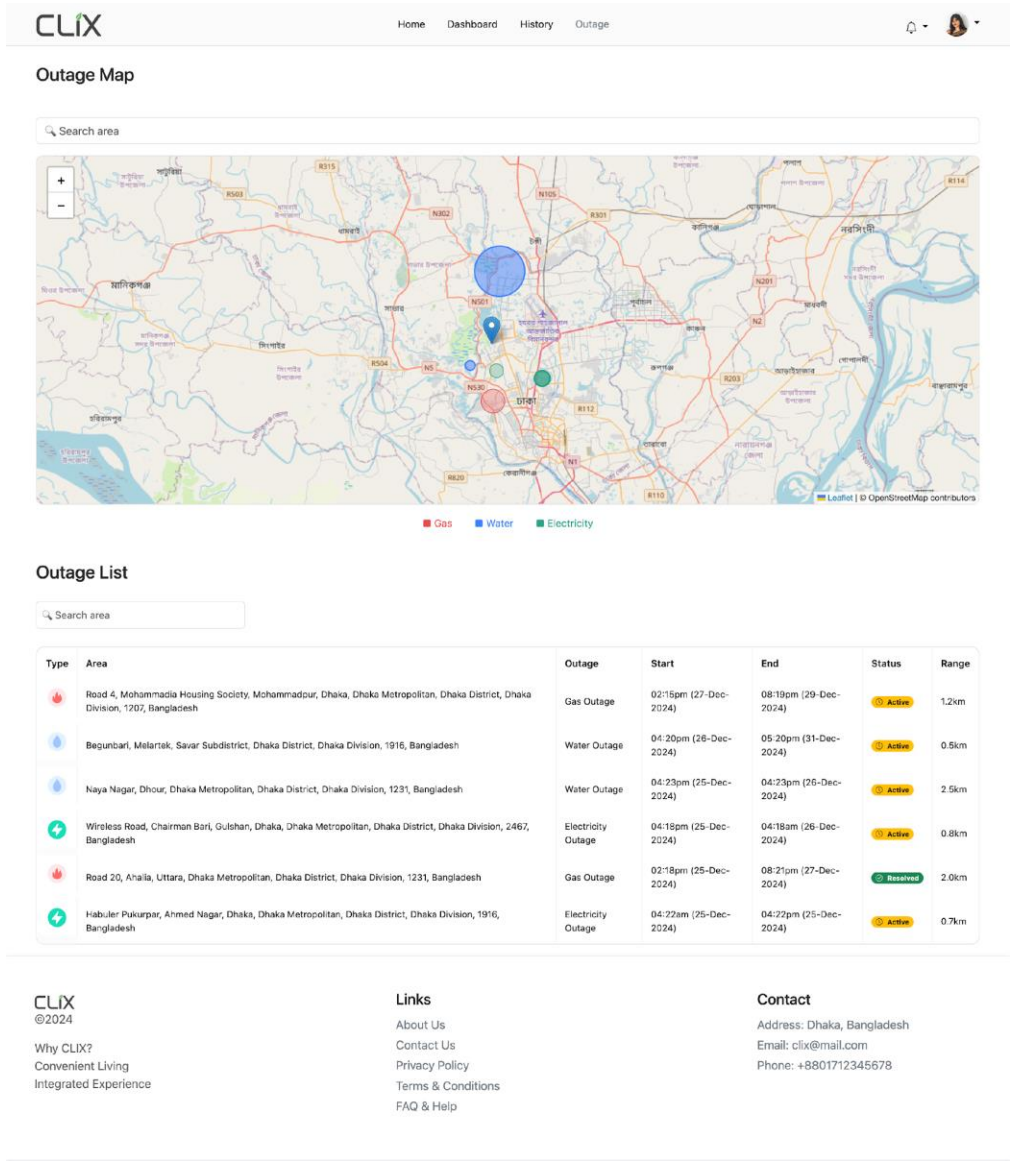
Links

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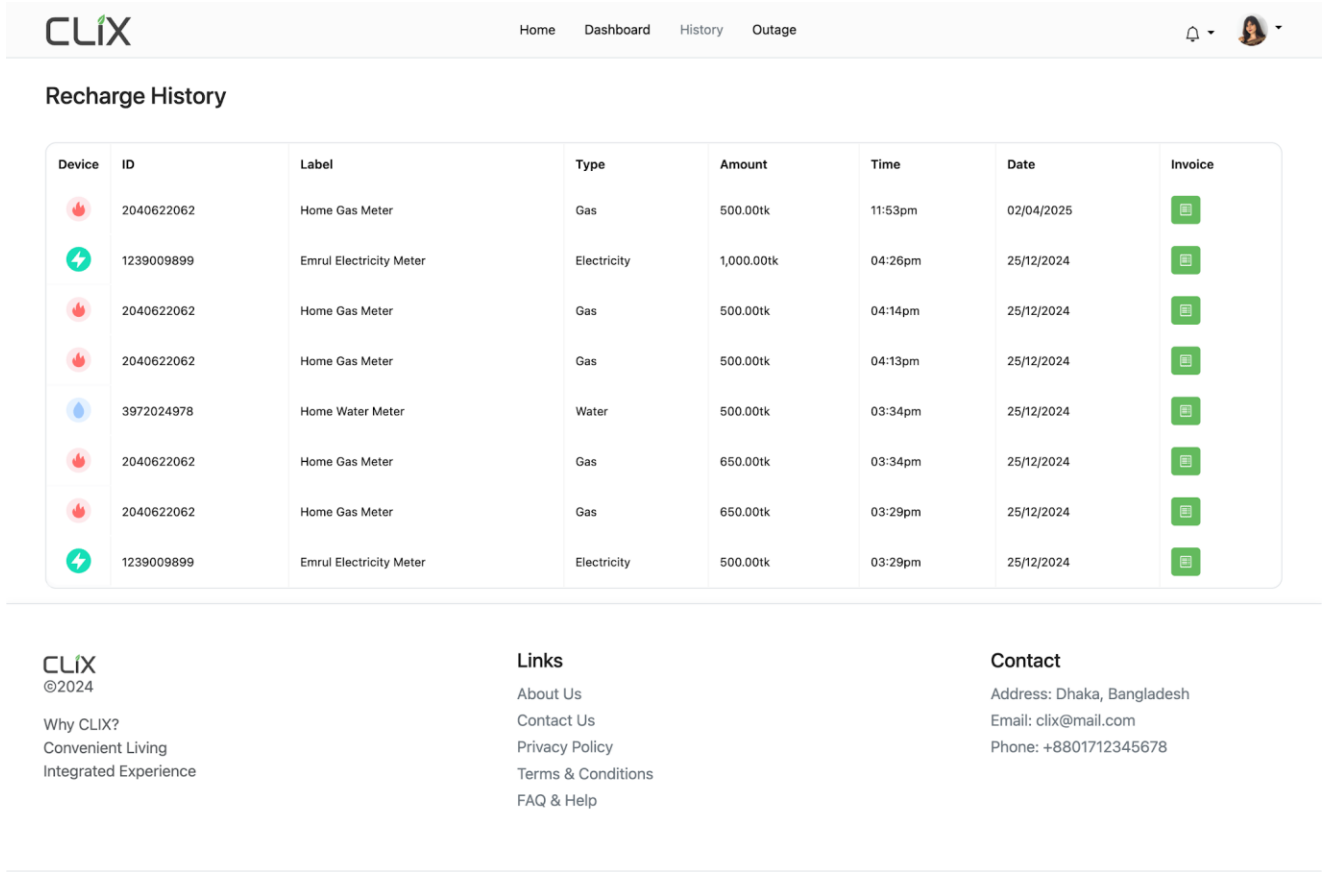
Contact

Address: Dhaka, Bangladesh
Email: clix@mail.com
Phone: +8801712345678

Dashboard


CLIX
Home
Dashboard
History
Outage
Outage Map
Search area
Leaflet | © OpenStreetMap contributors
Gas
Water
Electricity
Outage List
Search area
Type
Area
Outage
Start
End
Status
Range
Road 4, Mohammadia Housing Society, Mohammadpur, Dhaka, Dhaka Metropolitan, Dhaka District, Dhaka Division, 1207, Bangladesh
Gas Outage
02:15pm (27-Dec-2024)
08:19pm (29-Dec-2024)
Active
1.2km
Begunbari, Melartek, Savar Subdistrict, Dhaka District, Dhaka Division, 1916, Bangladesh
Water Outage
04:20pm (26-Dec-2024)
05:20pm (31-Dec-2024)
Active
0.5km
Naya Nagar, Dhour, Dhaka Metropolitan, Dhaka District, Dhaka Division, 1231, Bangladesh
Water Outage
04:23pm (25-Dec-2024)
04:23pm (26-Dec-2024)
Active
2.5km
Wireless Road, Chairman Bari, Gulshan, Dhaka, Dhaka Metropolitan, Dhaka District, Dhaka Division, 2467, Bangladesh
Electricity Outage
04:18pm (25-Dec-2024)
04:18am (26-Dec-2024)
Active
0.8km
Road 20, Ahalia, Uttara, Dhaka Metropolitan, Dhaka District, Dhaka Division, 1231, Bangladesh
Gas Outage
02:18pm (25-Dec-2024)
08:21pm (27-Dec-2024)
Received
2.0km
Habuler Pukurpar, Ahmed Nagar, Dhaka, Dhaka Metropolitan, Dhaka District, Dhaka Division, 1916, Bangladesh
Electricity Outage
04:22am (25-Dec-2024)
04:22pm (25-Dec-2024)
Active
0.7km
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Why CLIX?
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Integrated Experience
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Contact
Address: Dhaka, Bangladesh
Email: clix@mail.com
Phone: +8801712345678

Outage Map



Recharge History

CLIX Overview

- **Integrated Technology Platform**
  - **LoRaWAN-enabled smart meters** with tamper detection and **AES-128 encryption**
  - **Unified management dashboard** for real-time monitoring across all utilities
  - **AI-driven analytics** for:
    - Leak detection
    - Theft identification
    - Equipment failure prediction
  - **Multi-channel payment integration**, including mobile banking
  - **Real-time outage management** with push notifications
  - **Multi-language consumer portal** with usage tracking
-

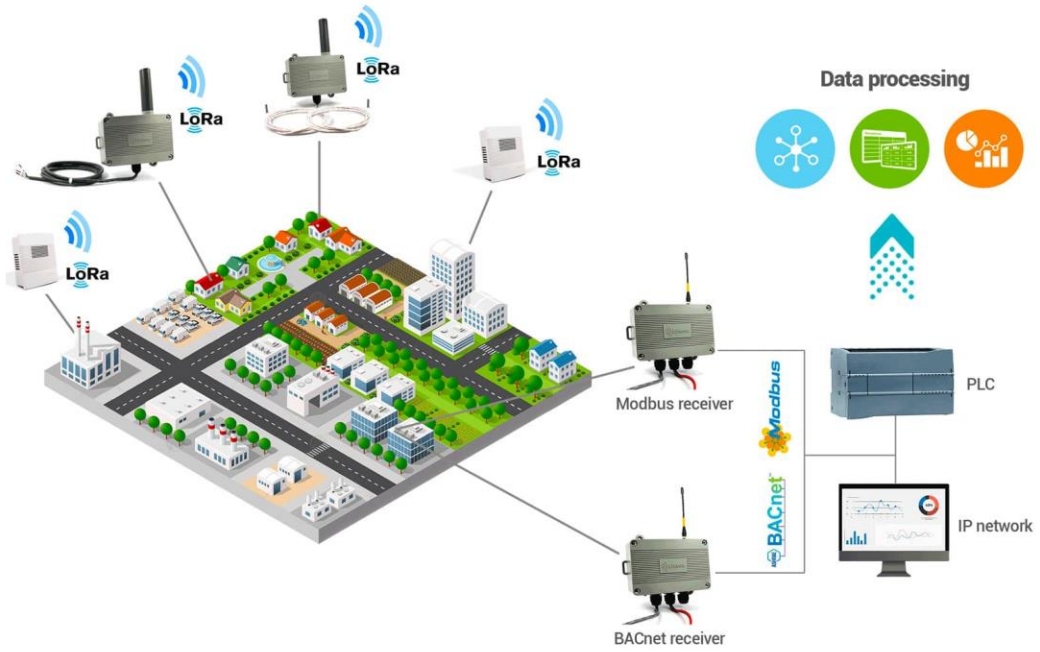
- **Key Objectives**
  - Reduce **non-revenue losses** across all utilities
  - Deploy **automated meter reading** and billing systems
  - Enhance service reliability via **predictive maintenance**
  - Establish a **unified platform** for all utility operations



Now What is AND Why LoRaWAN?

What is LoRaWAN?

- Long Range Wide Area Network technology for IoT applications
- Uses Chirp Spread Spectrum modulation for resilience
- Long-range connectivity (10-15km) with minimal power consumption
- Battery life of 7-10 years on a single charge
- AES-128 encryption for secure end-to-end data protection
- Operation on license-free spectrum, eliminating ongoing costs
- Support for thousands of devices per gateway with minimal infrastructure







Technical Advantages of LoRaWAN

Current Challenges

- **5.72%** losses in electricity. (DESCO Annual Report 2023)
- Limited effectiveness of current GSM-based communication systems

Comparison with Current Systems:

- **Coverage:** 10-15 km range vs. 1-2 km for GSM
- **Battery Life:** 7-10 years vs. 2-3 years for GSM-based systems
- **Infrastructure:** One gateway serves thousands of devices
- **Operating Costs:** No ongoing SIM card or cellular data costs
- **Security:** End-to-end encryption with AES-128

Global Evidence for LoRaWAN Effectiveness

- **Indonesia:** Singapore-based Sindcon is retrofitting 50,000+ smart meters in Jakarta with STM32WLE5 LoRaWAN microcontrollers, enhancing remote meter reading in diverse environments while providing 10-year battery life. (The GPS Time, 2023)
- **Singapore:** Sindcon achieved 90%+ data success rates and up to 7-year battery life with their LoRaWAN smart meters, significantly improving utility management efficiency. (Smart Energy, 2025)
- **Switzerland:** Elvexys and Oiken implemented LoRa-enabled sensors for real-time power grid fault detection, reducing deployment costs by 90% compared to traditional methods. (Semtech, 2021)
- **France:** CAHORS Group's Sentinel® line fault detection solutions use LoRaWAN to monitor power lines, enabling immediate fault identification and preventing costly outages. (Semtech, 2019)



Proven Evidence of LoRa Success in Bangladesh

Chattogram WASA Pilot:

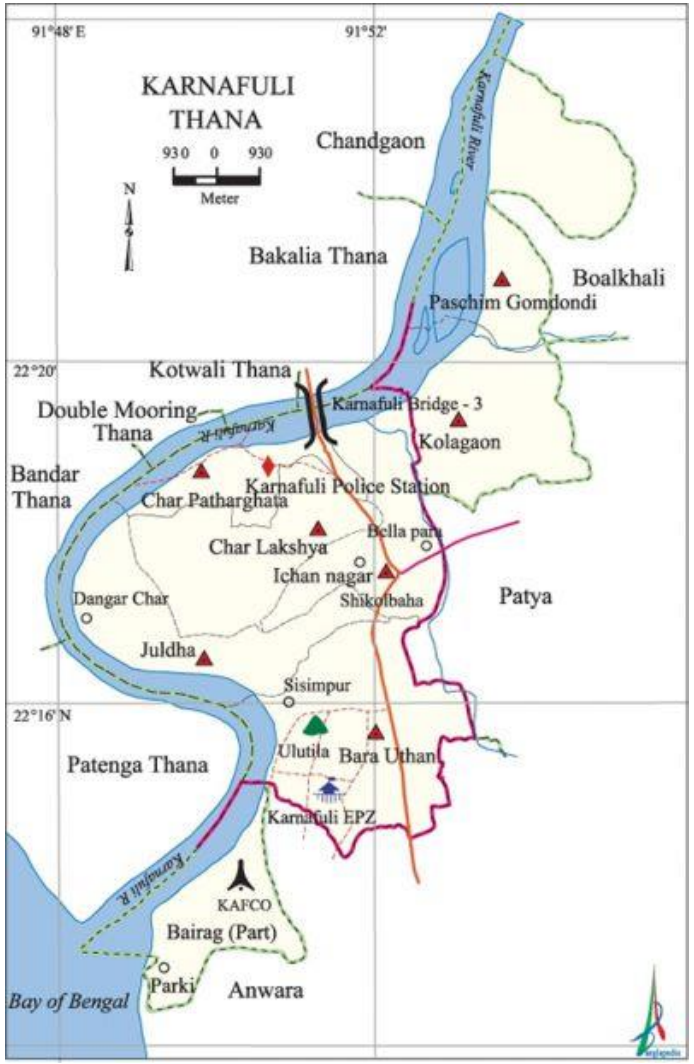
- "3,000 Pure Ultrasonic Digital meters (LoRa)" already implemented (The Business Standard, 2022)
- "With digital meters no one can tamper with it nor can steal. Also all the bills will automatically come to their server." (Md Samsul Alam, Deputy Managing Director, Chittagong Wasa – Independent TV)

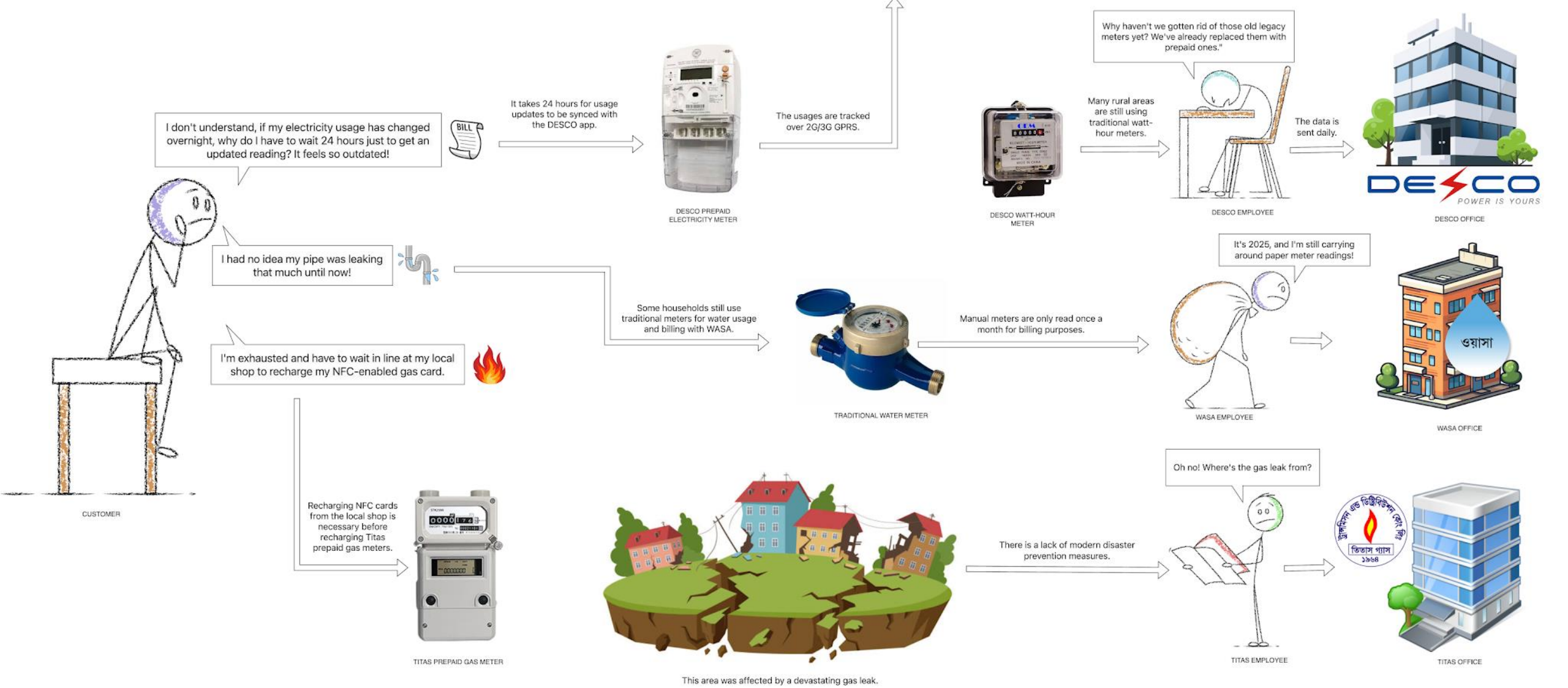
Karnaphuli Water Supply Project:

- Successfully "reduced non-revenue water to 10%" (The Business Standard, 2024)
- "The Karnaphuli Service Area... with DMA system providing uninterrupted supply to 46,000 connections" (The Business Standard, 2024)

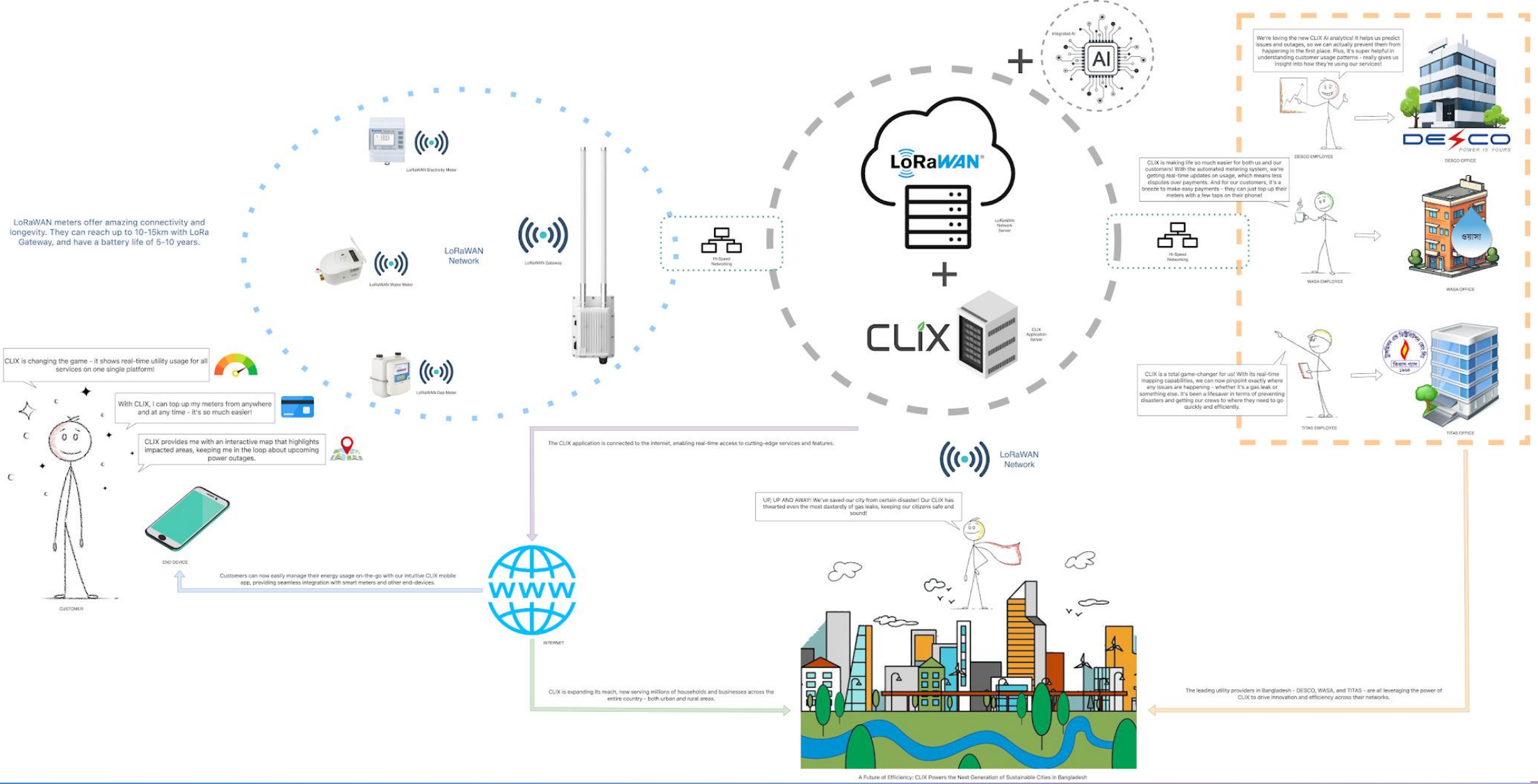
WASA Official Assessment:

- "Hopefully, if the project is successfully implemented, water wastage will be prevented and Chattogram Wasa's revenue will increase." (Maksud Alam, Chief Engineer, The Business Standard, 2022)




I don't understand, if my electricity usage has changed overnight, why do I have to wait 24 hours just to get an updated reading? It feels so outdated!
BILL
It takes 24 hours for usage updates to be synced with the DESCO app.
DESCO PREPAID ELECTRICITY METER
The usages are tracked over 2G/3G GPRS.
DESCO WATT-HOUR METER
Many rural areas are still using traditional watt-hour meters.
Why haven't we gotten rid of those old legacy meters yet? We've already replaced them with prepaid ones."
DESCO EMPLOYEE
The data is sent daily.
DESCO
POWER IS YOURS
DESCO OFFICE
I had no idea my pipe was leaking that much until now!
Some households still use traditional meters for water usage and billing with WASA.
TRADITIONAL WATER METER
Manual meters are only read once a month for billing purposes.
It's 2025, and I'm still carrying around paper meter readings!
WASA EMPLOYEE
ওয়াসা
WASA OFFICE
I'm exhausted and have to wait in line at my local shop to recharge my NFC-enabled gas card.
CUSTOMER
Recharging NFC cards from the local shop is necessary before recharging Titas prepaid gas meters.
TITAS PREPAID GAS METER
This area was affected by a devastating gas leak.
There is a lack of modern disaster prevention measures.
Oh no! Where's the gas leak from?
TITAS EMPLOYEE
তিতাস গ্যাস
১৯৬৪
TITAS OFFICE

As Is System


LoRaWAN meters offer amazing connectivity and longevity. They can reach up to 10-15km with LoRa Gateway, and have a battery life of 5-10 years.
LoRaWAN Network
LoRaWAN Gateway
Hi-Speed Networking
LoRaWAN Network Server
CLIX Application Server
AI
CLIX is changing the game - it shows real-time utility usage for all services on one single platform!
With CLIX, I can top up my meters from anywhere and at any time - it's so much easier!
CLIX provides me with an interactive map that highlights impacted areas, keeping me in the loop about upcoming power outages.
END DEVICE
CUSTOMER
Customers can now easily manage their energy usage on-the-go with our intuitive CLIX mobile app, providing seamless integration with smart meters and other end-devices.
INTERNET
The CLIX application is connected to the internet, enabling real-time access to cutting-edge services and features.
CLIX is expanding its reach, now serving millions of households and businesses across the entire country - both urban and rural areas.
LoRaWAN Network
UP, UP AND AWAY! We've saved our city from certain disaster! Our CLIX has thwarted even the most dastardly of gas leaks, keeping our citizens safe and sound!
We're loving the new CLIX AI analytics! It helps us predict issues and outages, so we can actually prevent them from happening in the first place. Plus, it's super helpful in understanding customer usage patterns - really gives us insight into how they're using our services!
DESCO EMPLOYEE
DESCO OFFICE
CLIX is making life so much easier for both us and our customers! With the automated metering system, we're getting real-time updates on usage, which means less disputes over payments. And for our customers, it's a breeze to make easy payments - they can just top up their meters with a few taps on their phone!
WASA EMPLOYEE
WASA OFFICE
CLIX is a total game-changer for us! With its real-time mapping capabilities, we can now pinpoint exactly where any issues are happening - whether it's a gas leak or something else. It's been a lifesaver in terms of preventing disasters and getting our crews to where they need to go quickly and efficiently.
TITAS EMPLOYEE
TITAS OFFICE
The leading utility providers in Bangladesh - DESCO, WASA, and TITAS - are all leveraging the power of CLIX to drive innovation and efficiency across their networks.
A Future of Efficiency: CLIX Powers the Next Generation of Sustainable Cities in Bangladesh

To Be

Market Size & Opportunity

Target Market:

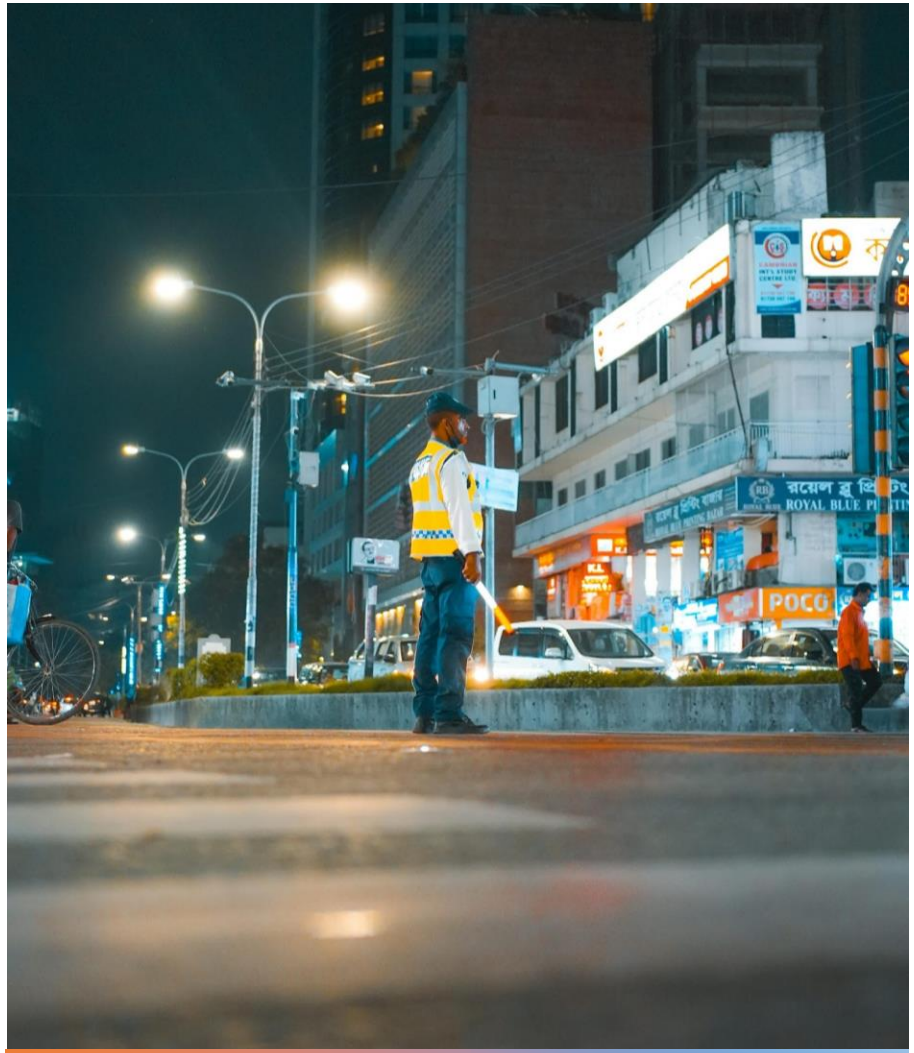
- 40M+ households across Bangladesh
- 500K+ commercial and industrial entities

Addressing Critical Infrastructure:

- Water supply networks serving major urban centers
- Electricity distribution covering 3.5 crore connections
- Gas utility systems with documented 7.67% losses

Corruption Elimination:

- Evidence that "meter inspectors in collusion with WASA officials show less reading and provide illegal connections in exchange for illegal financial benefits" (The Business Standard, 2022)
- CLIX removes human intervention points that enable corruption



Feature4	Highly Helpful	Helpful	Neutral	Not Helpful	Not Helpful at all
Emergency gas leak alert at night	88.60%	10.10%	0.00%	0.00%	1.30%
Real-time electricity bill tracking	67.10%	29.10%	3.80%	0.00%	0.00%
Water leak alert when away from home	70.90%	27.80%	1.30%	0.00%	0.00%
One-click bill payment via mobile	74.70%	24.10%	1.30%	0.00%	0.00%
Preventing payment for others' illegal connections	74.70%	24.10%	1.30%	0.00%	0.00%
Load-shedding timing notifications	75.90%	20.30%	3.80%	0.00%	0.00%
Integrated system for electricity, gas, water	65.80%	27.80%	6.30%	0.00%	0.00%
Tech & job opportunities from smart systems	72.20%	24.10%	3.80%	0.00%	0.00%
Accurate billing with usage record	75.90%	17.10%	6.30%	0.00%	0.00%
Disaster-time live utility map	70.90%	25.3%	3.80%	0.00%	0.00%
Usage-based saving recommendations	69.60%	24.10%	5.10%	1.30%	0.00%
Adoption of global smart utility models	73.40%	13.90%	12.7%	0.00%	0.00%
Prior notification of utility personnel arrival	65.80%	22.80%	10.10%	1.30%	0.00%
Knowing which appliances use the most electricity	67.10%	31.60%	1.30%	0.00%	0.00%
Leaving reliable systems for future generations	68.40%	22.80%	8.90%	0.00%	0.00%
Average	**72.07%**	**23.00%**	**4.65%**	**0.17%**	**0.09%**

Survey Analytics

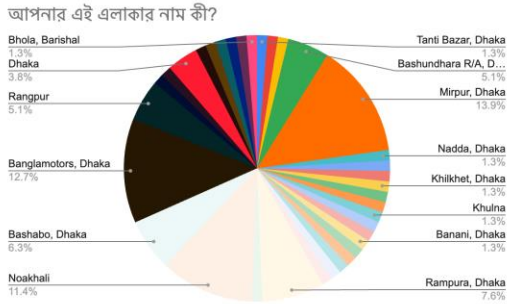
*We collected **79** surveys **(26 online, 53 offline)**. CLIX received **"Highly Helpful"** reviews from **72.07%** of respondents, **"Helpful"** from **23.00%**, **"Neutral"** from **4.65%**, **"Not Helpful"** from **0.17%**, and **"Not Helpful at all" 0.09%** of respondents.*

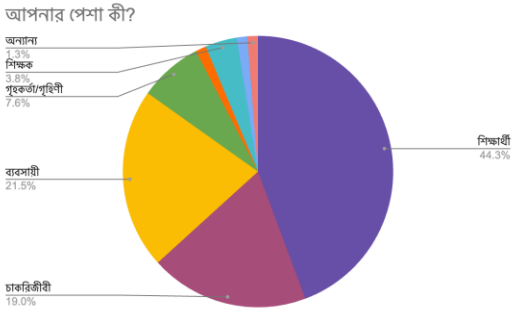
Survey Analytics

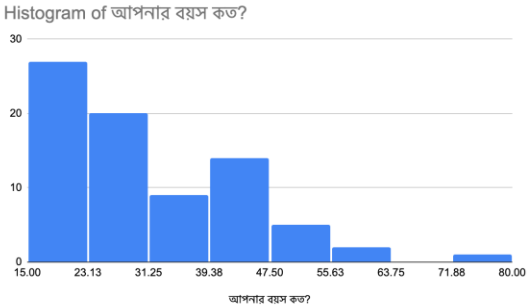
Occupation: 44.3% Student, 21.5% Businessman, 19.0% Service holder, 7.6% Householder, 3.8% Teacher & 3.8% Other

Area: Dhaka = 54.4% & Outside Dhaka = 45.6%

Age: 18-25: 35.4% (28 people), 26-35: 25.3% (20 people), 36-45: 19.0% (15 people), 46-55: 12.7% (10 people), 56-65: 6.3% (5 people) & 66+: 1.3% (1 person)







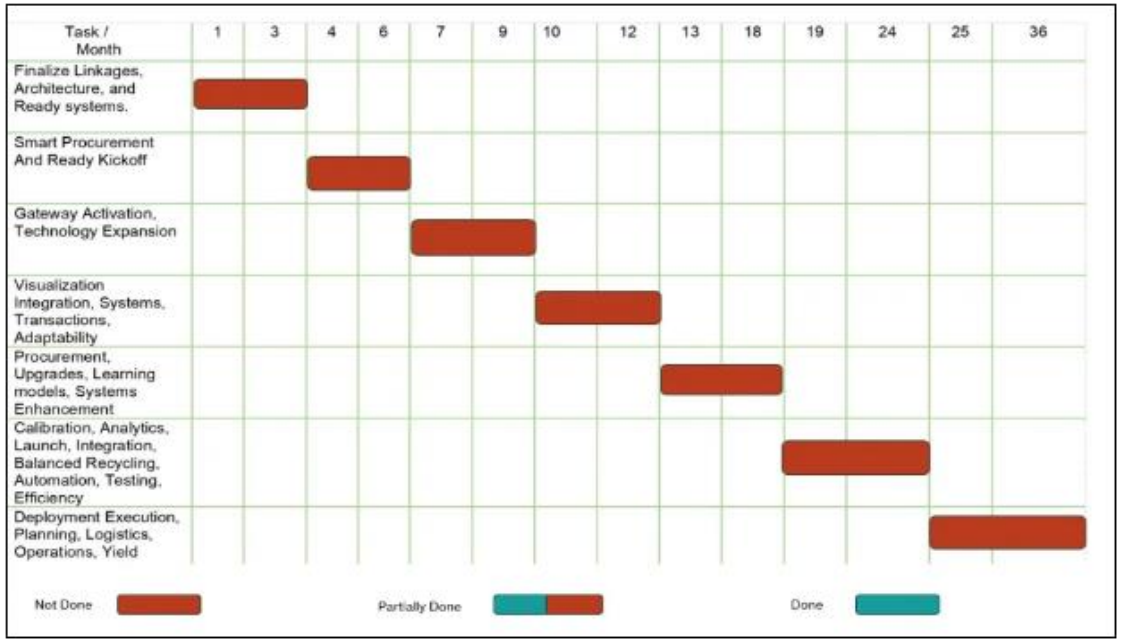
Implementation Plan

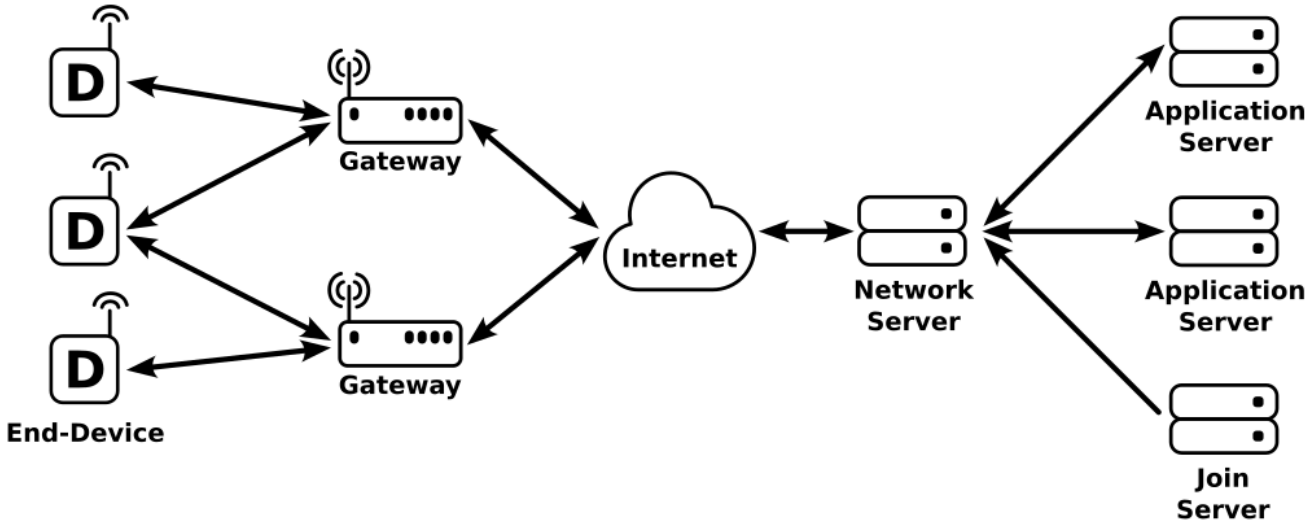
Comprehensive Project Lifecycle (36 months):

- Research Phase (3 months): Needs assessment, regulatory compliance
- Design Phase (4 months): System architecture, UI/UX, security protocols
- Development Phase (6 months): Core components, integrations, analytics
- Testing Phase (3 months): Quality assurance, security validation
- Pilot Deployment (3 months): 5,000 households in selected urban area
- Full Deployment (12 months): Phased rollout across regions
- Maintenance & Enhancement (5 months): Ongoing support

Key Milestones:

- Urban Deployment (Month 25): 100,000+ connections in major cities
- Full Deployment (Month 31): Nationwide coverage
- System Integration (Month 33): Complete integration with utilities
- Performance Optimization (Month 36): System refinement





System Architecture

Core Components:

- **LoRaWAN-enabled Smart Meters:**
  - Tamper-resistant design with real-time monitoring
  - AES-128 encryption for security
  - Battery life of 7-10 years
  - Examples include models like STK25MI gas meters
- **LoRaWAN Network Infrastructure:**
  - Carrier-grade gateways with 10-15 km range
  - Hybrid connectivity options for reliability
  - Mesh network capabilities for urban environments
- **Centralized Management Platform:**
  - Real-time dashboard for utility monitoring
  - Outage mapping and management
  - Payment integration system
  - Analytics engine for pattern detection
- **User Interfaces:**
  - Mobile application for consumers
  - Web portal for utility authorities
  - API integration for third-party services

Comparative Analysis

- **Current System vs. CLIX System:**
- **Key Improvements:**
- Non-revenue water loss reduction from 25-30% to 10%
- Elimination of 5.72% electricity distribution losses
- Minimization of 7.67% gas system inefficiencies
- 95% detection accuracy for unusual consumption vs. <30% in current systems
- Average payment time reduced from 45 minutes to under 2 minutes

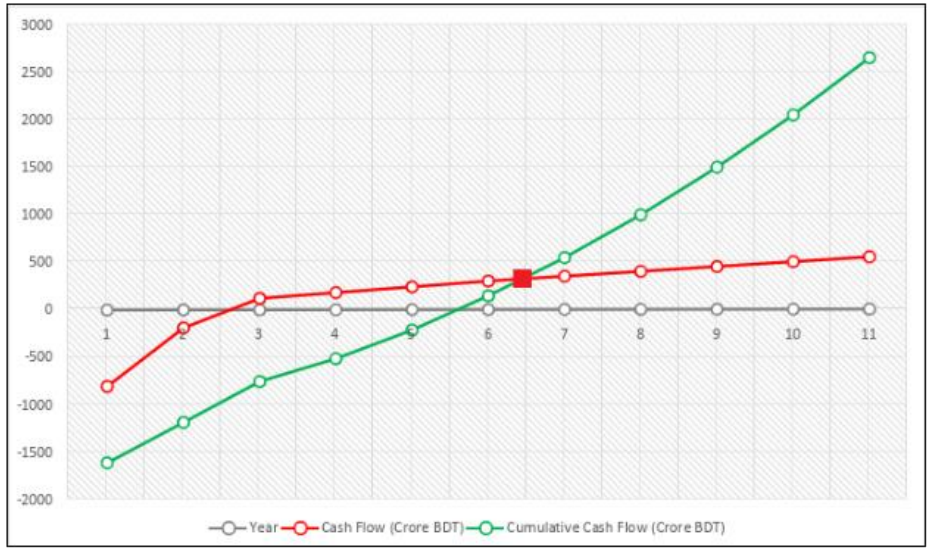
Feature	Current System ("As Is")	CLIX System ("To Be")
Meter Reading	Manual, once per month	Automated, real-time
Usage Updates	24-hour delay	Instant monitoring
Fraud Prevention	Vulnerable to collusion	Tamper-resistant design
Payment System	Physical visits required	Remote mobile payment
Consumer Visibility	Limited insight	Detailed analytics
System Integration	Siloed utilities	Unified platform

REVENUE Model

Customer-Based Revenue: Utility providers will share a portion of their revenue with CLIX.

Enterprise-Based Revenue: CLIX will provide advanced monitoring systems to enterprise users. These users will pay utility providers, and CLIX will receive a share of this revenue. Additionally, CLIX will earn income through subscription fees.





Economic Feasibility

Investment Overview:

- Total project cost: 1420.55 crore BDT
- Initial investment: 566.5 crore BDT
- 10-year maintenance costs: 675.0 crore BDT
- Contingency (10%): 129.05 crore BDT

Financial Metrics:

- Net Present Value (NPV): 448.73 crore BDT
- ROI: 87.95%
- Payback Period: 5.44 years (5 years, 5 months)
- Total Cash Inflow over 10 years: 2670 crore BDT
- Net Benefit: 1249.45 crore BDT



SUSTAINABILITY

Economic Benefits:

- 50% reduction in manual meter-reading expenses
- 15-20% reduction in utility waste (20-25 crore BDT annually)
- 50-60 crore BDT in annual energy cost reductions

Environmental Benefits:

- CO_2 emission reduction: 1.5 million tons per year
- Water conservation: 2-3 billion liters annually

Social Benefits:

- Job creation: 20,000-50,000 new positions
- 60% reduction in unplanned outage duration
- 40% fewer billing disputes



Why Invest in CLIX

- Building on proven success of Chattogram WASA's smart meter pilot
- Comprehensive implementation plan with clear milestones
- Multiple revenue streams ensuring long-term sustainability

Key Impact Quote: "We are losing more than Tk45 crore in revenue every year [just in Chattogram]. WASA officials hope that digital meters can prevent this wastage and corruption." (The Business Standard, 2022)

Thank You

"The people who are crazy enough to think they can change the world are the ones who do." - Steve Jobs

